

EXT 1: Calculus (Ext1), C3 Applications of Calculus (Y12), Differential Equations (Ext1)

Equations and Slope Fields

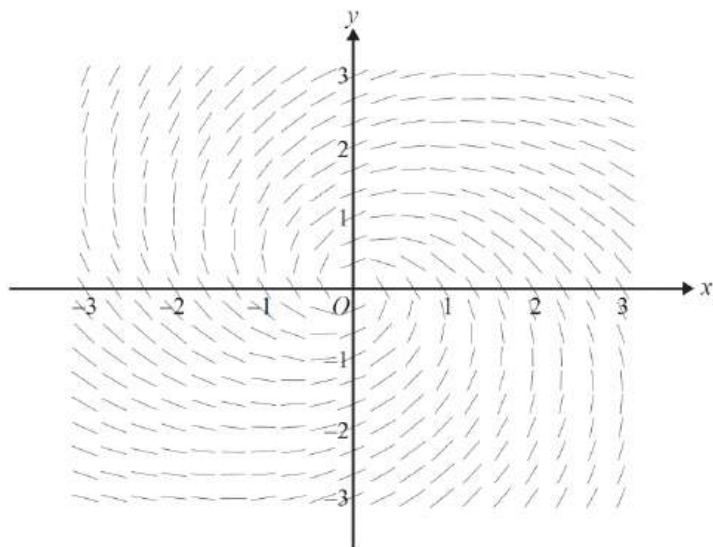
Applications of Differential Equations

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Exam Equivalent Time: 27 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Questions

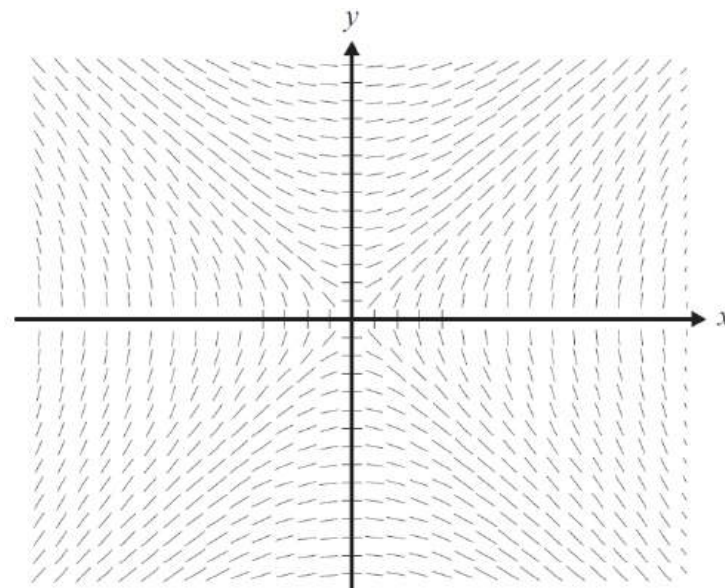
1. Calculus, EXT1 C3 2011 SPEC2 17 MC



The differential equation which best represents the above direction field is

- A. $\frac{dy}{dx} = \frac{y - 2x}{2y + x}$
- B. $\frac{dy}{dx} = \frac{2x - y}{y - 2x}$
- C. $\frac{dy}{dx} = \frac{2y - x}{y + 2x}$
- D. $\frac{dy}{dx} = \frac{y - 2x}{2y - x}$
- E. $\frac{dy}{dx} = \frac{x - 2y}{2y + x}$

2. Calculus, EXT1 C3 2013 SPEC2 12 MC



The differential equation that best represents the above direction field is

- A. $\frac{dy}{dx} = x^2 - y^2$
- B. $\frac{dy}{dx} = y^2 - x^2$
- C. $\frac{dy}{dx} = -\frac{x}{y}$
- D. $\frac{dy}{dx} = \frac{x}{y}$

3. Calculus, EXT1 C3 2013 VCE 13 MC

Water containing 2 grams of salt per litre flows at the rate of 10 litres per minute into a tank that initially contained 50 litres of pure water. The concentration of salt in the tank is kept uniform by stirring and the mixture flows out of the tank at the rate of 6 litres per minute.

If Q grams is the amount of salt in the tank t minutes after the water begins to flow, the differential equation relating Q to t is

- A. $\frac{dQ}{dt} = 20 - \frac{3Q}{25 + 2t}$
 B. $\frac{dQ}{dt} = 10 - \frac{3Q}{25 + 2t}$
 C. $\frac{dQ}{dt} = 20 - \frac{3Q}{25 - 2t}$
 D. $\frac{dQ}{dt} = 10 - \frac{3Q}{25 - 2t}$

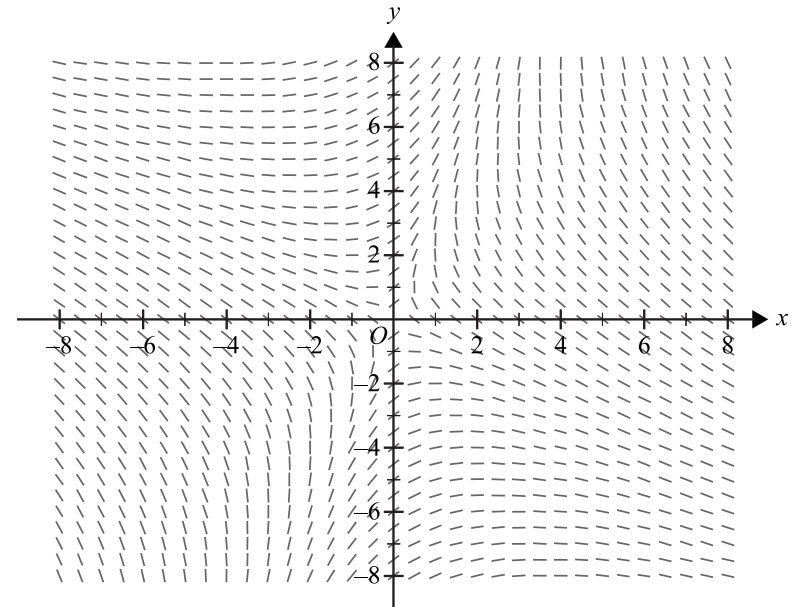
4. Calculus, EXT1 C3 2014 VCE 10 MC

A large tank initially holds 1500 L of water in which 100 kg of salt is dissolved. A solution containing 2 kg of salt per litre flows into the tank at a rate of 8 L per minute. The mixture is stirred continuously and flows out of the tank through a hole at a rate of 10 L per minute.

The differential equation for Q , the number of kilograms of salt in the tank after t minutes, is given by

- A. $\frac{dQ}{dt} = 16 - \frac{5Q}{750 - t}$
 B. $\frac{dQ}{dt} = 16 - \frac{5Q}{750 + t}$
 C. $\frac{dQ}{dt} = 16 + \frac{5Q}{750 - t}$
 D. $\frac{dQ}{dt} = \frac{100Q}{750 - t}$

5. Calculus, EXT1 C3 2018 SPEC2 10 MC



The differential equation that best represents the direction field above is

- A. $\frac{dy}{dx} = \frac{2x + y}{y - 2x}$
 B. $\frac{dy}{dx} = \frac{x + 2y}{2x - y}$
 C. $\frac{dy}{dx} = \frac{2x - y}{x + 2y}$
 D. $\frac{dy}{dx} = \frac{x - 2y}{y - 2x}$

6. Calculus, EXT1 C3 2015 SPEC2 12

Find y given $\frac{dy}{dx} = 1 - \frac{y}{3}$ and $y = 4$ when $x = 2$. (2 marks)

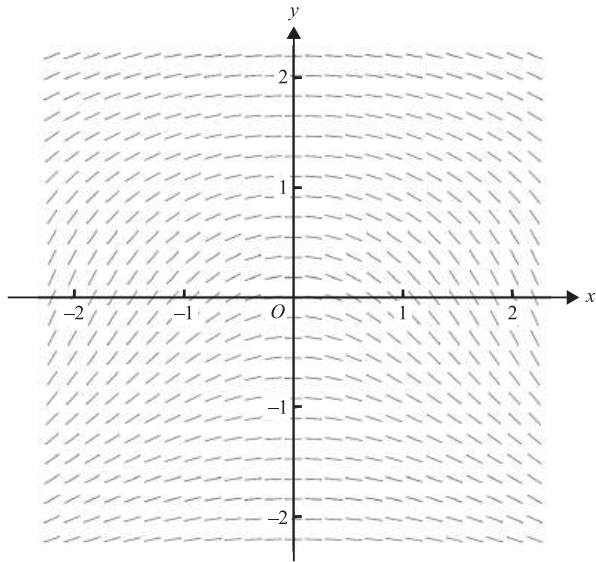
7. Calculus, EXT1 C3 2017 SPEC1-N 7

Let $\frac{dy}{dx} = (4 - y)^2$.

Express y in terms of x , where $y(0) = 3$. (3 marks)

8. Calculus, EXT1 C3 2017 SPEC1 8

A slope field representing the differential equation $\frac{dy}{dx} = -\frac{x}{1+y^2}$ is shown below.



- Sketch the solution curve of the differential equation corresponding to the condition $y(-1) = 1$ on the slope field above and, hence, estimate the positive value of x when $y = 0$. Give your answer correct to one decimal place. (2 marks)
- Solve the differential equation $\frac{dy}{dx} = \frac{-x}{1+y^2}$ with the condition $y(-1) = 1$. Express your answer in the form $ay^3 + by + cx^2 + d = 0$, where a, b, c and d are integers. (2 marks)

9. Calculus, EXT1 C3 2016 SPEC1 10

Solve the differential equation $\sqrt{2-x^2} \frac{dy}{dx} = \frac{1}{2-y}$, given that $y(1) = 0$. Express y as a function of x . (4 marks)

Worked Solutions

1. Calculus, EXT1 C3 2011 SPEC2 17 MC

When $x = 0 \Rightarrow$ gradients are all positive

Eliminate B and E.

When $y = 0 \Rightarrow$ gradients are all negative

Eliminate D.

Option A will have zero gradient along $y = 2x$ (correct)

Option C will have zero gradient along $y = \frac{1}{2}x$ (incorrect)

$\Rightarrow A$

Almost half of all students answered incorrectly - mean mark 52%.

2. Calculus, EXT1 C3 2013 SPEC2 12 MC

By inspection:

When $x = 0 \Rightarrow \frac{dy}{dx} = 0$

When $y = 0 \Rightarrow \frac{dy}{dx} \rightarrow \infty$

$\therefore$ Eliminate A, and B

Along $y = x \Rightarrow \frac{dy}{dx} > 0$

$\therefore$ Eliminate C

$\Rightarrow D$

3. Calculus, EXT1 C3 2013 VCE 13 MC

$$\begin{aligned}\text{Volume} &= 50 + (10 - 6)t \\ &= 50 + 4t\end{aligned}$$

Salt in tank at time $t = Q$ grams

$$\therefore \text{Concentration} = \frac{Q}{50 + 4t} \text{ grams per litre}$$

$$\frac{dQ}{dt}_{\text{in}} = 2 \times 10 = 20 \text{ g/min}$$

$$\begin{aligned}\frac{dQ}{dt}_{\text{out}} &= 6 \times \frac{Q}{50 + 4t} \\ &= \frac{3Q}{25 + 2t}\end{aligned}$$

$$\begin{aligned}\therefore \frac{dQ}{dt} &= 20 - \frac{3Q}{25 + 2t} \\ &\Rightarrow A\end{aligned}$$

4. Calculus, EXT1 C3 2014 VCE 10 MC

$$\frac{dQ_{\text{in}}}{dV} = 2 \text{ kg/L}, \quad \frac{dV_{\text{in}}}{dt} = 8 \text{ L/min}$$

$$V_0 = 1500, \quad Q_0 = 100$$

$$\frac{dV_{\text{out}}}{dt} = 10 \text{ L/min}$$

$$\begin{aligned}V(t) &= 1500 + (8 - 10)t \\ &= 1500 - 2t \\ &= 2(750 - t)\end{aligned}$$

$$\frac{dQ_{\text{in}}}{dt} = 2 \times 8 = 16 \text{ kg/min}$$

$$\begin{aligned}\frac{dQ_{\text{out}}}{dt} &= \frac{Q}{v(t)} \times 10 \\ &= \frac{10Q}{2(750 - t)} \\ &= \frac{5Q}{750 - t}\end{aligned}$$

$$\begin{aligned}\therefore \frac{dQ}{dt} &= 16 - \frac{5Q}{750 - t} \\ &\Rightarrow A\end{aligned}$$

5. Calculus, EXT1 C3 2018 SPEC2 10 MC

When $x = 0$, $m = 1$

When $y = 0$, $m = -1$

$\Rightarrow A$

6. Calculus, EXT1 C3 2015 SPEC2 12

$$\frac{dy}{dx} = \frac{3-y}{3}$$

$$\frac{dx}{dy} = \frac{3}{3-y}$$

$$x = \int \frac{3}{3-y} dy$$

$$\frac{x}{3} = -\ln|3-y| + c$$

Given $y = 4$ when $x = 2$:

$$\frac{2}{3} = -\ln|-1| + c$$

$$c = \frac{2}{3}$$

$$\frac{x}{3} = -\ln|3-y| + \frac{2}{3}$$

$$\ln|3-y| = \frac{2-x}{3}$$

$$3-y = \pm e^{\frac{2-x}{3}}$$

$$\therefore y = 3 + e^{\frac{2-x}{3}}$$

7. Calculus, EXT1 C3 2017 SPEC1-N 7

$$\frac{dy}{dx} = (4-y)^2$$

$$\frac{dx}{dy} = \frac{1}{(4-y)^2}$$

$$x = \int \frac{1}{(4-y)^2} dy$$

$$= \int (4-y)^{-2} dy$$

$$= (-1)(-1)(4-y)^{-1} + c$$

$$= \frac{1}{4-y} + c$$

When $x = 0$, $y = 3$:

$$0 = \frac{1}{4-3} + c$$

$$\therefore c = -1$$

$$x = \frac{1}{4-y} - 1$$

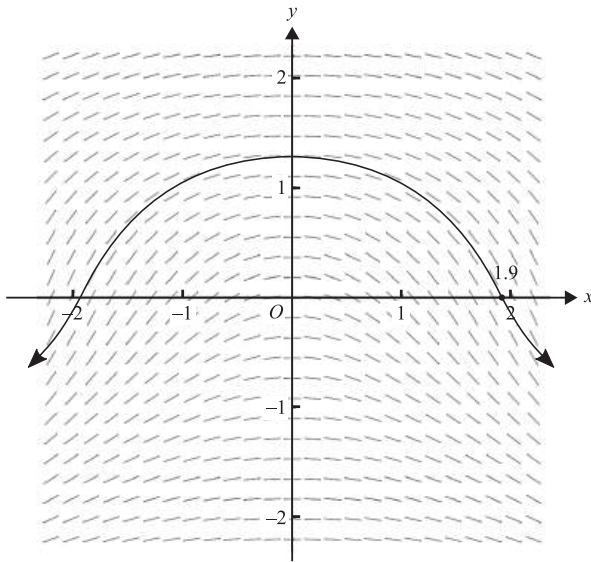
$$x+1 = \frac{1}{4-y}$$

$$\frac{1}{x+1} = 4-y$$

$$\therefore y = 4 - \frac{1}{x+1}$$

8. Calculus, EXT1 C3 2017 SPEC1 8

a.



♦♦ Mean mark part (a) 32%.
MARKER'S COMMENT: Solution curve should follow slope ticks and not cross them.

b. $(1 + y^2) \frac{dy}{dx} = -x$

$$\int 1 + y^2 dy = - \int x dx$$

$$y + \frac{y^3}{3} = -\frac{x^2}{2} + C, C \in \mathbb{R}$$

Substituting $(-1, 1)$:

$$1 + \frac{1^3}{3} = -\frac{(-1)^2}{2} + C$$

$$1 + \frac{1}{3} = -\frac{1}{2} + C$$

$$\therefore C = \frac{11}{6}$$

$$y + \frac{1}{3}y^3 = -\frac{1}{2}x^2 + \frac{11}{6}$$

$$6y + 2y^3 = -3x^2 + 11$$

$$\therefore 2y^3 + 6y + 3x^2 - 11 = 0$$

9. Calculus, EXT1 C3 2016 SPEC1 10

$$\sqrt{2 - x^2} \cdot \frac{dy}{dx} = \frac{1}{2 - y}$$

$$(2 - y) \cdot \frac{dy}{dx} = \frac{1}{\sqrt{2 - x^2}}$$

$$\int 2 - y dy = \int \frac{1}{\sqrt{2 - x^2}} dx$$

$$2y - \frac{y^2}{2} = \sin^{-1}\left(\frac{x}{\sqrt{2}}\right) + c$$

Given $y(1) = 0$:

$$0 = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + c$$

$$c = -\frac{\pi}{4}$$

$$2y - \frac{y^2}{2} = \sin^{-1}\left(\frac{x}{\sqrt{2}}\right) - \frac{\pi}{4}$$

$$y^2 - 4y = -2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right) + \frac{\pi}{2}$$

$$(y - 2)^2 - 4 = -2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right) + \frac{\pi}{2}$$

$$(y - 2)^2 = 4 + \frac{\pi}{2} - 2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)$$

$$(y - 2) = \pm \sqrt{4 + \frac{\pi}{2} - 2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)}$$

$$y = 2 \pm \sqrt{4 + \frac{\pi}{2} - 2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)}$$

Given $y = 0$ when $x = 1$:

$$\therefore y = 2 - \sqrt{4 + \frac{\pi}{2} - 2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)}$$

♦ Mean mark 46%.